Open Source Summit North America 2024

Sharing Reset GPIOs in the Linux Kernel

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Introduction

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- I work for Linaro in Qualcomm Landing Team
- I am the co-maintainer (with Rob and Conor) of Devicetree bindings in the Linux kernel
  - I also maintain other Linux kernel pieces
    - Memory controller drivers, NFC, 1-Wire subsystem
    - Samsung Exynos SoC ARM/ARM64 architecture
      - Do you know it covers also Google Tensor SoC in Google Pixel 6 phone?
  - I am also one of most active contributors to the Linux kernel

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- GPIO Aggregator and delay idea
- Maybe we need GPIO-enable-counting?
- Talk focuses on the Devicetree use cases (was submitted to EOSS)
  - Few solutions are Devicetree specific
  - There will be snippets of Devicetree Sources code
  - I don’t know how ACPI is handling this
    - Probably does not handle

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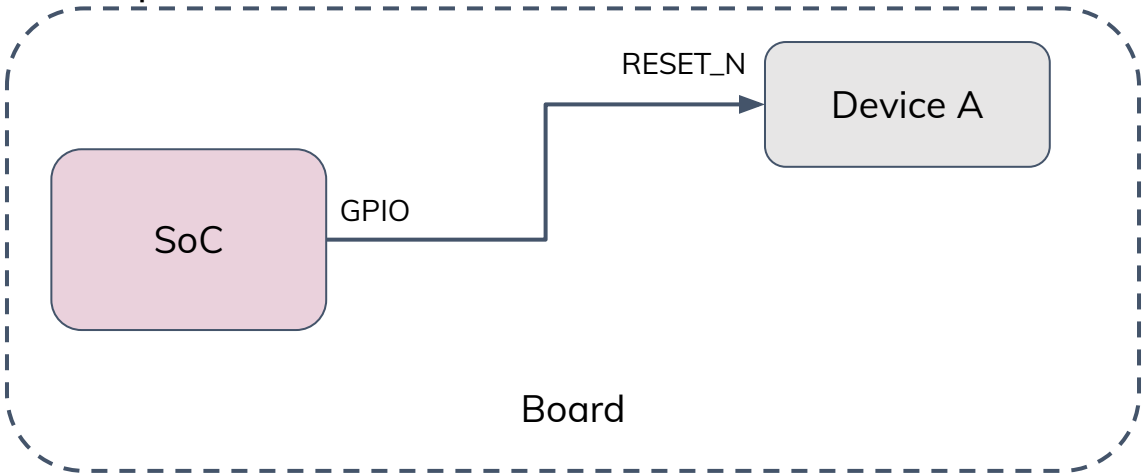

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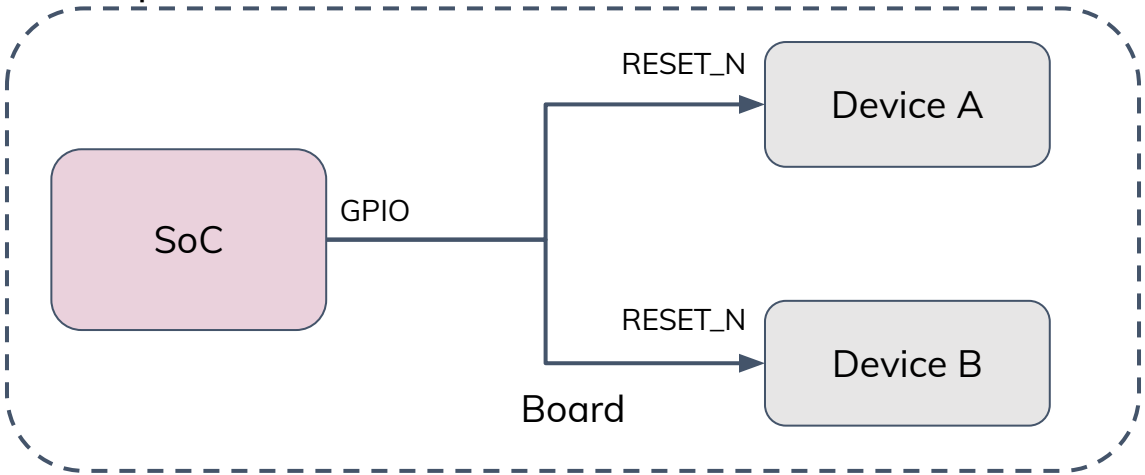
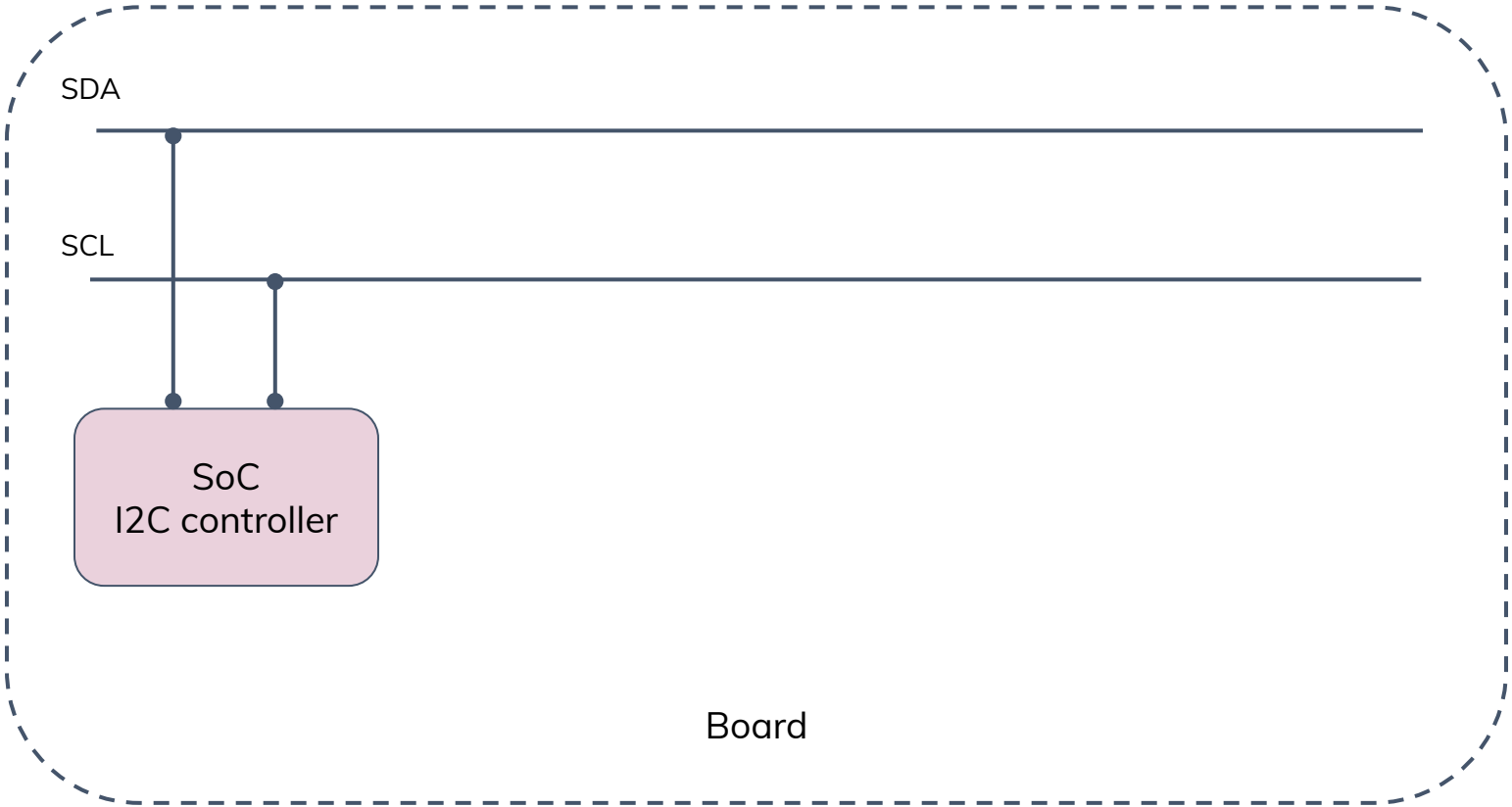


Example: I2C



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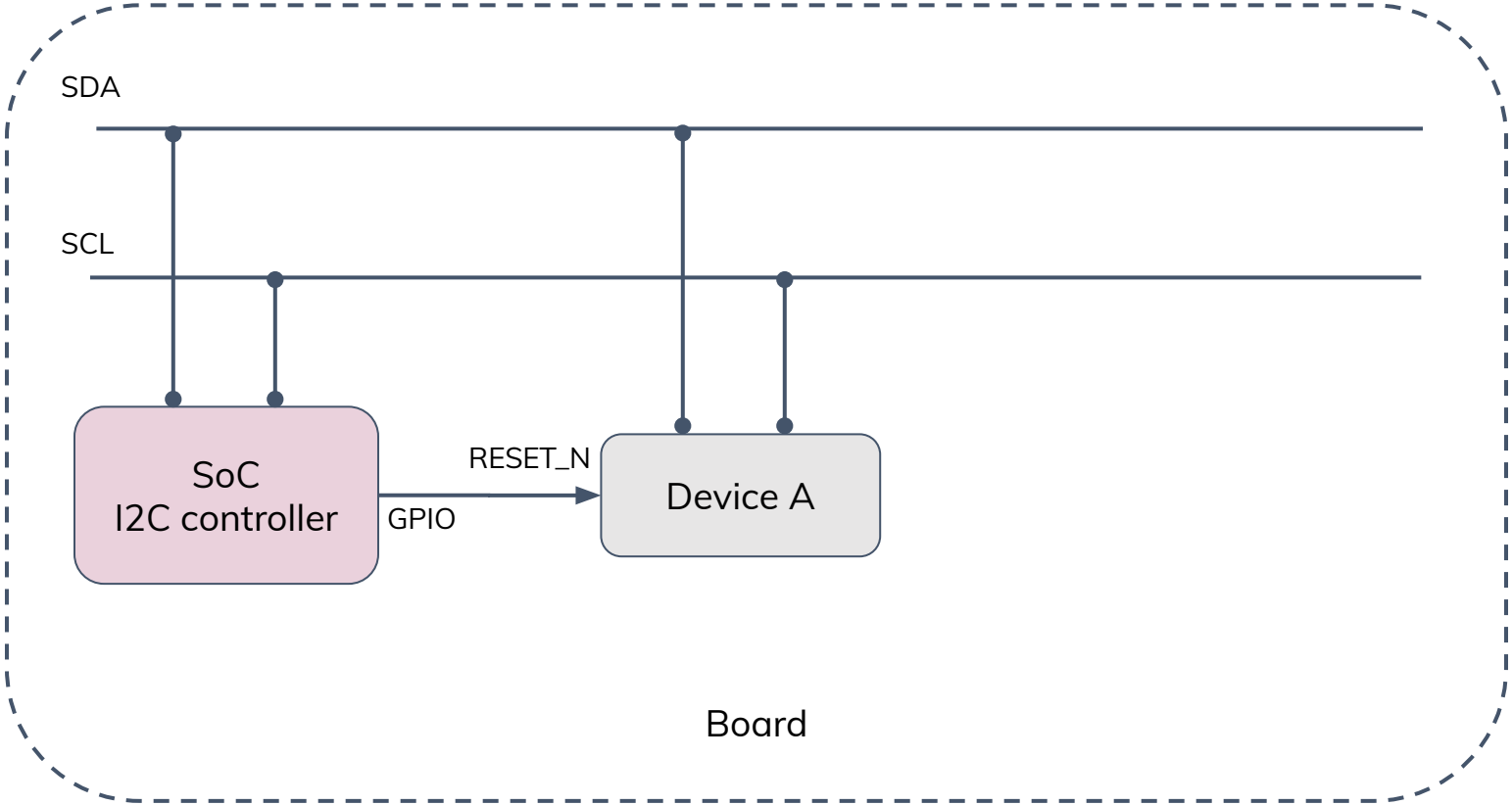

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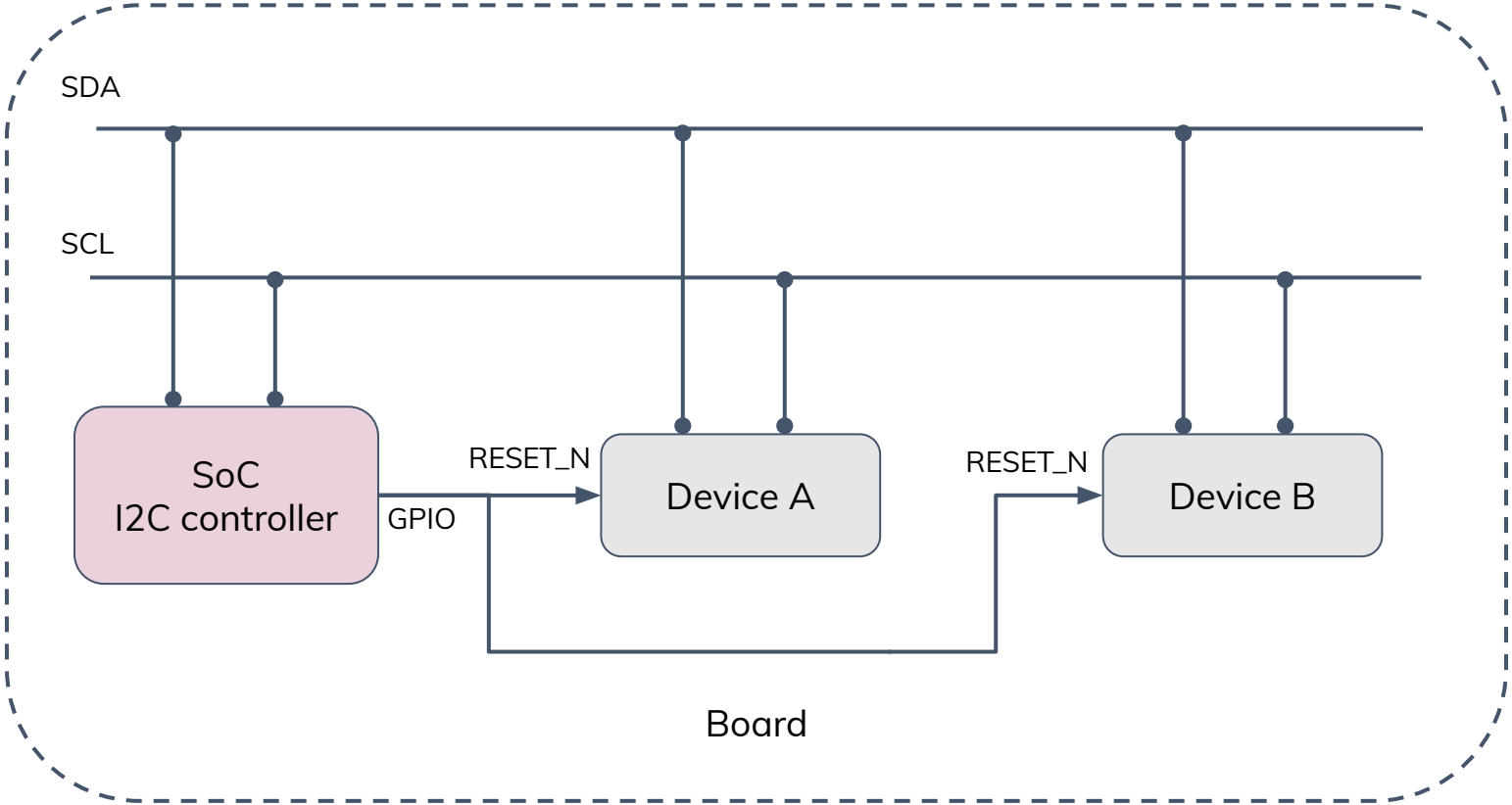
SDA
SCL
SoC
I2C controller
GPIO
RESET_N
Device A
RESET_N
Device B
Board

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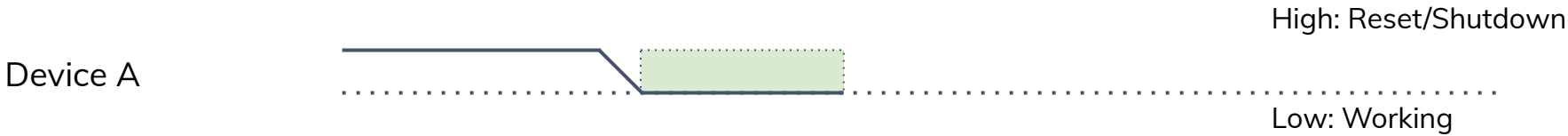
- When reset is asserted, all devices will go to reset
- Same the other way: de-assertion will turn on all devices

Shared Reset or Shutdown Line, Active High



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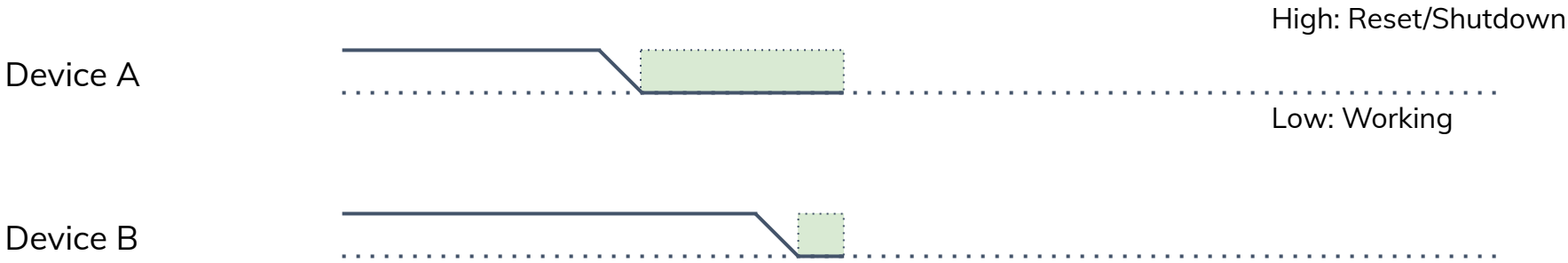

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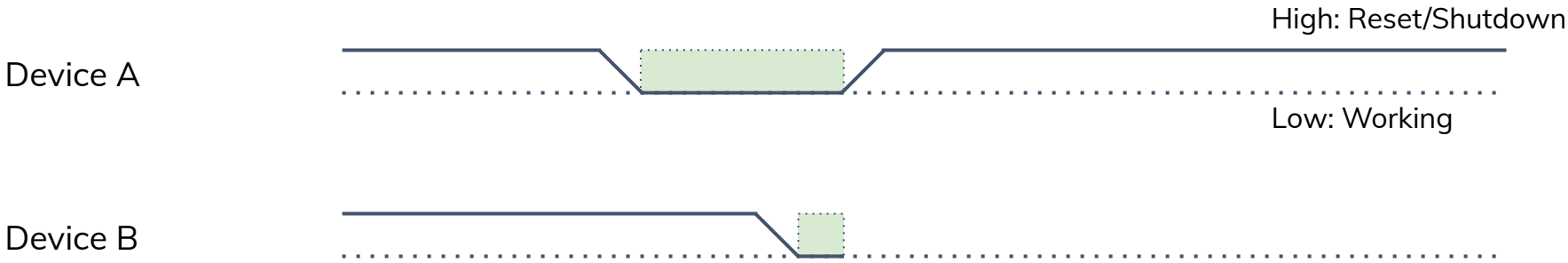

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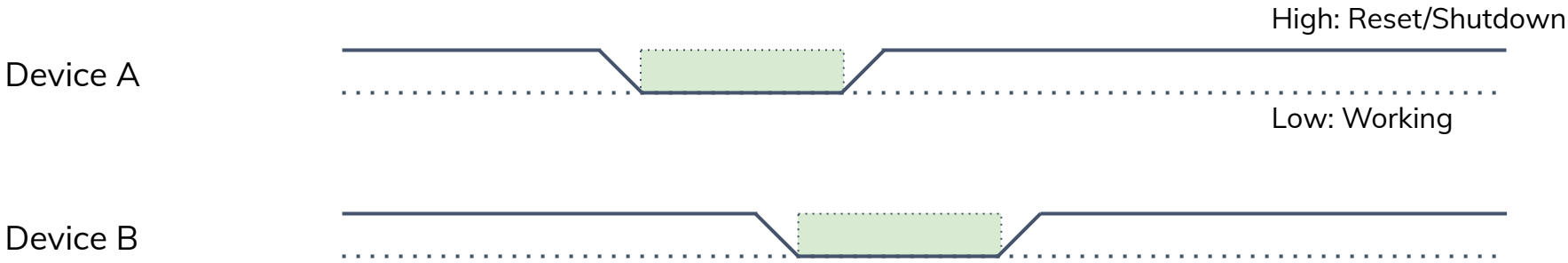

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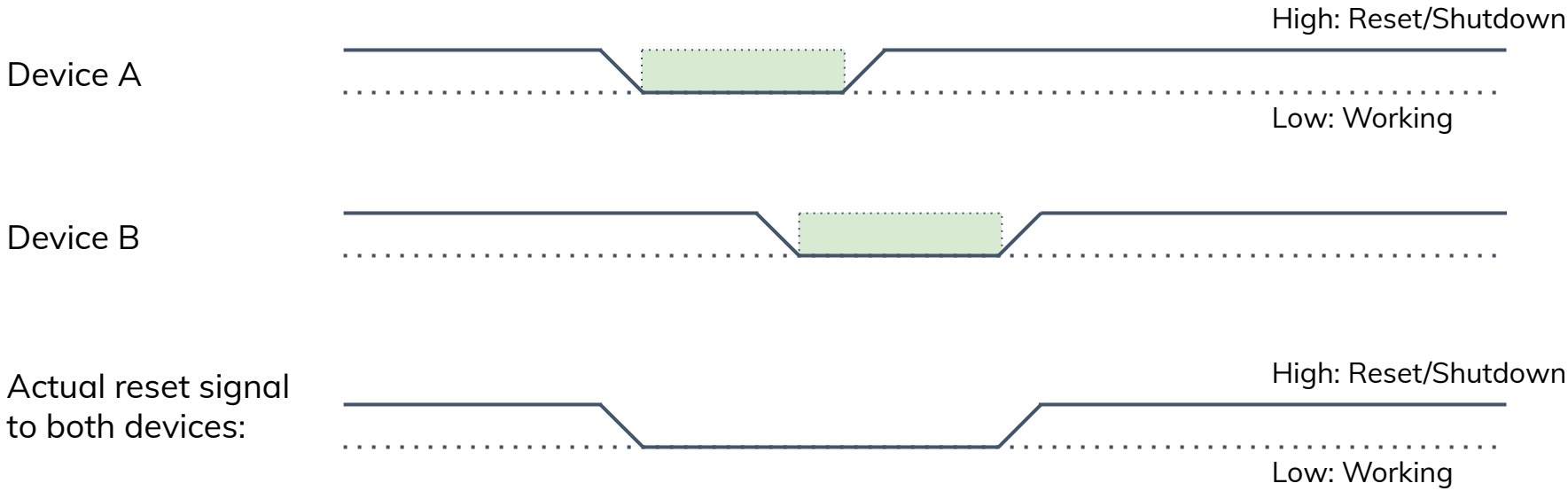

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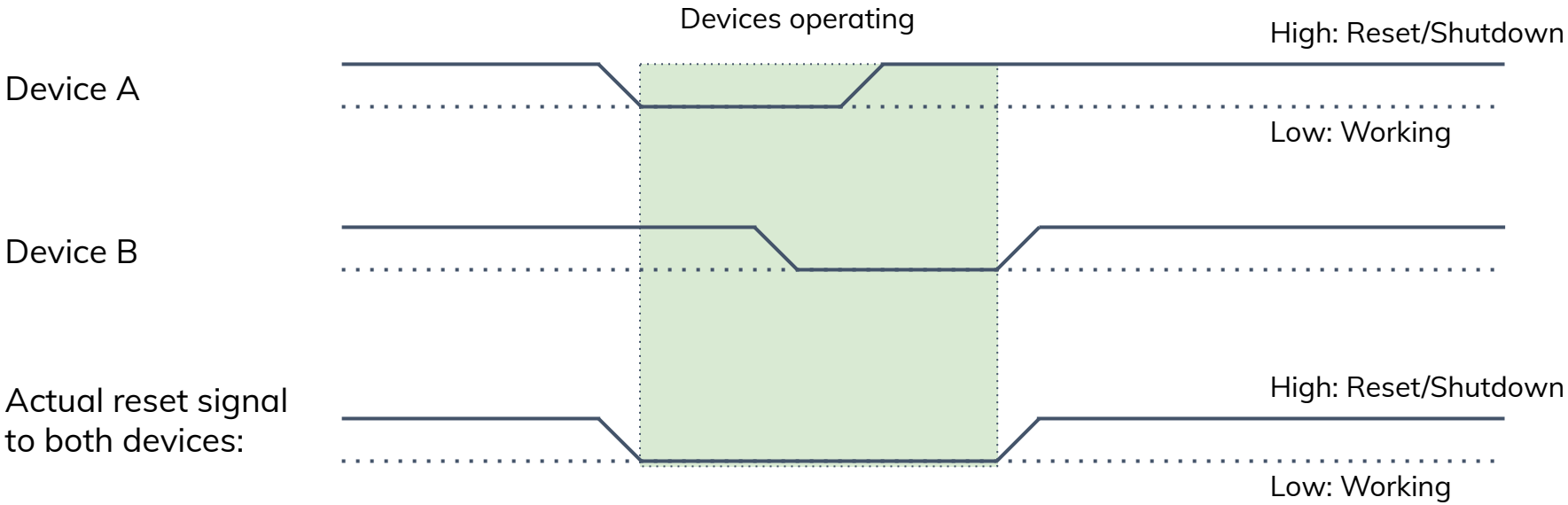

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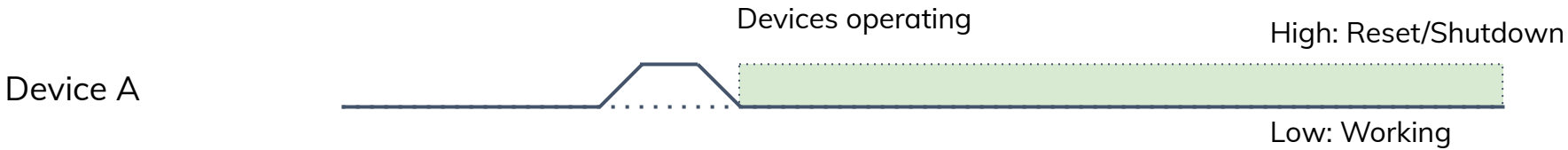

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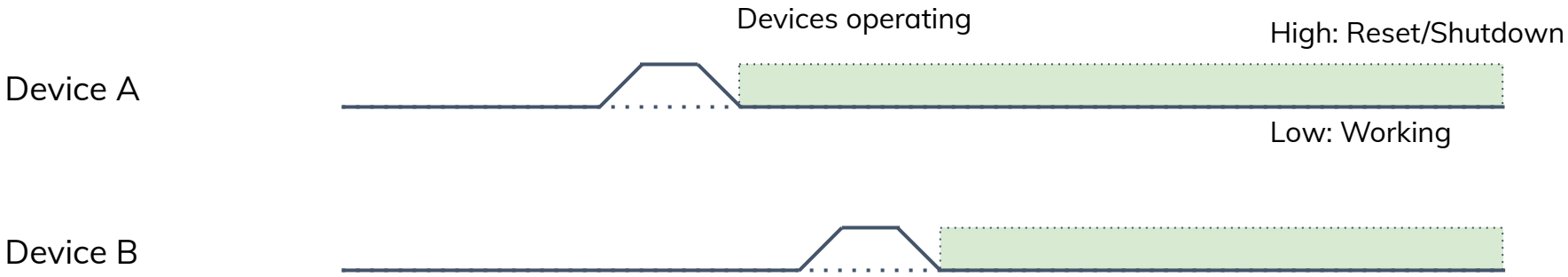

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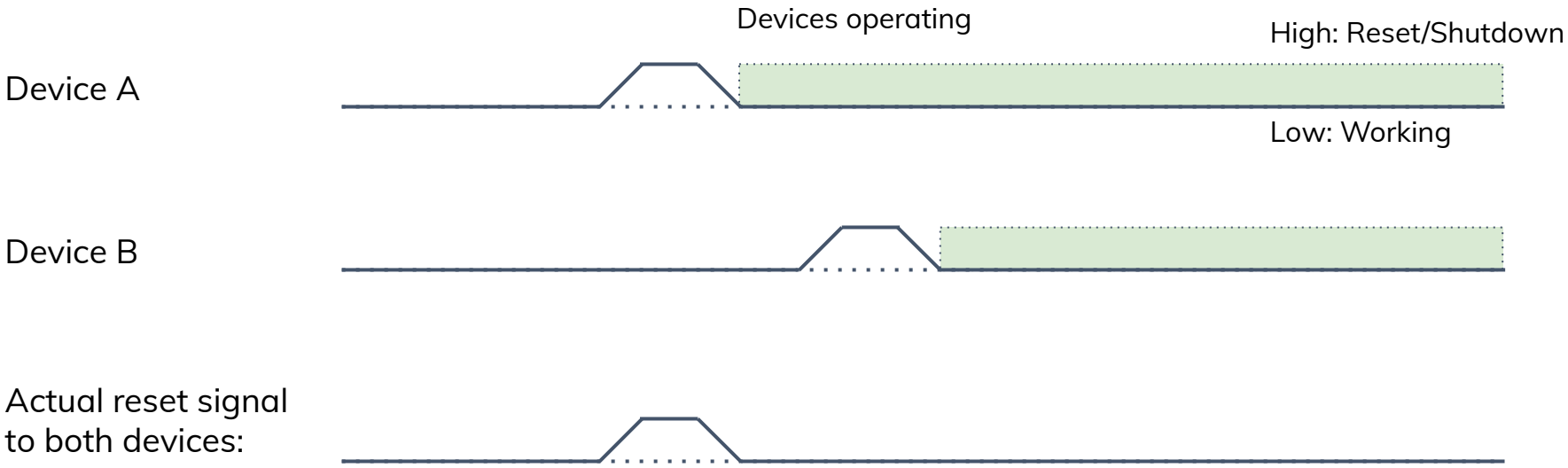

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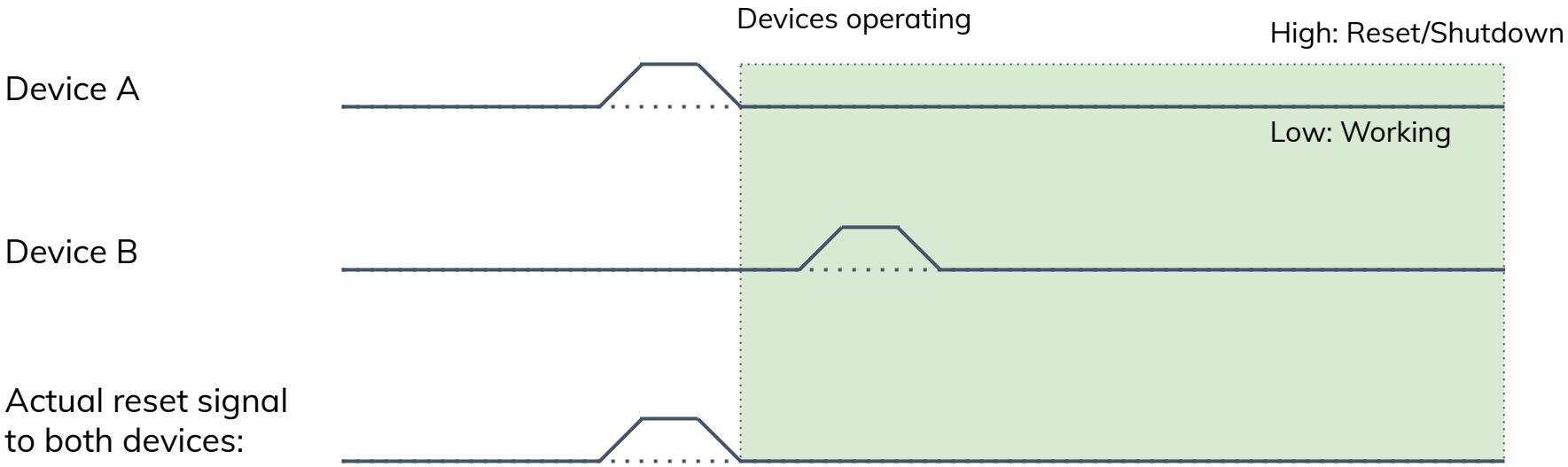

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- Linux GPIO framework does not even allow multiple drivers to request the same GPIO
  - More on that later

(Simplified) Linux Device Driver Probing

Bus match

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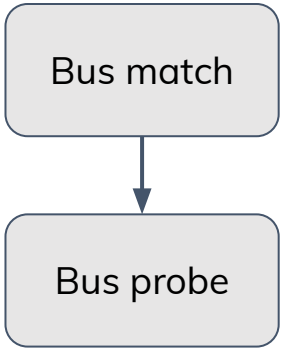

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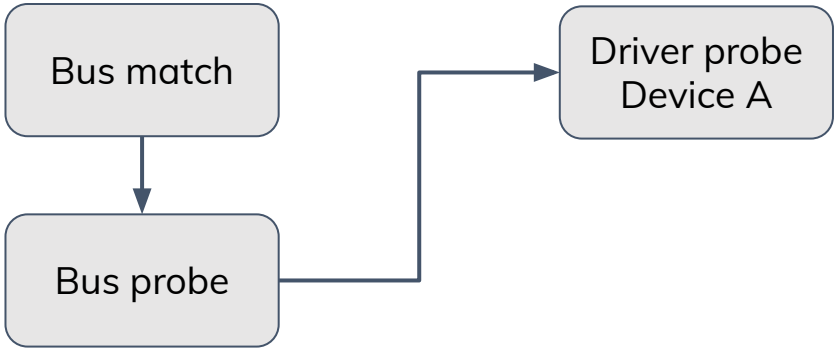

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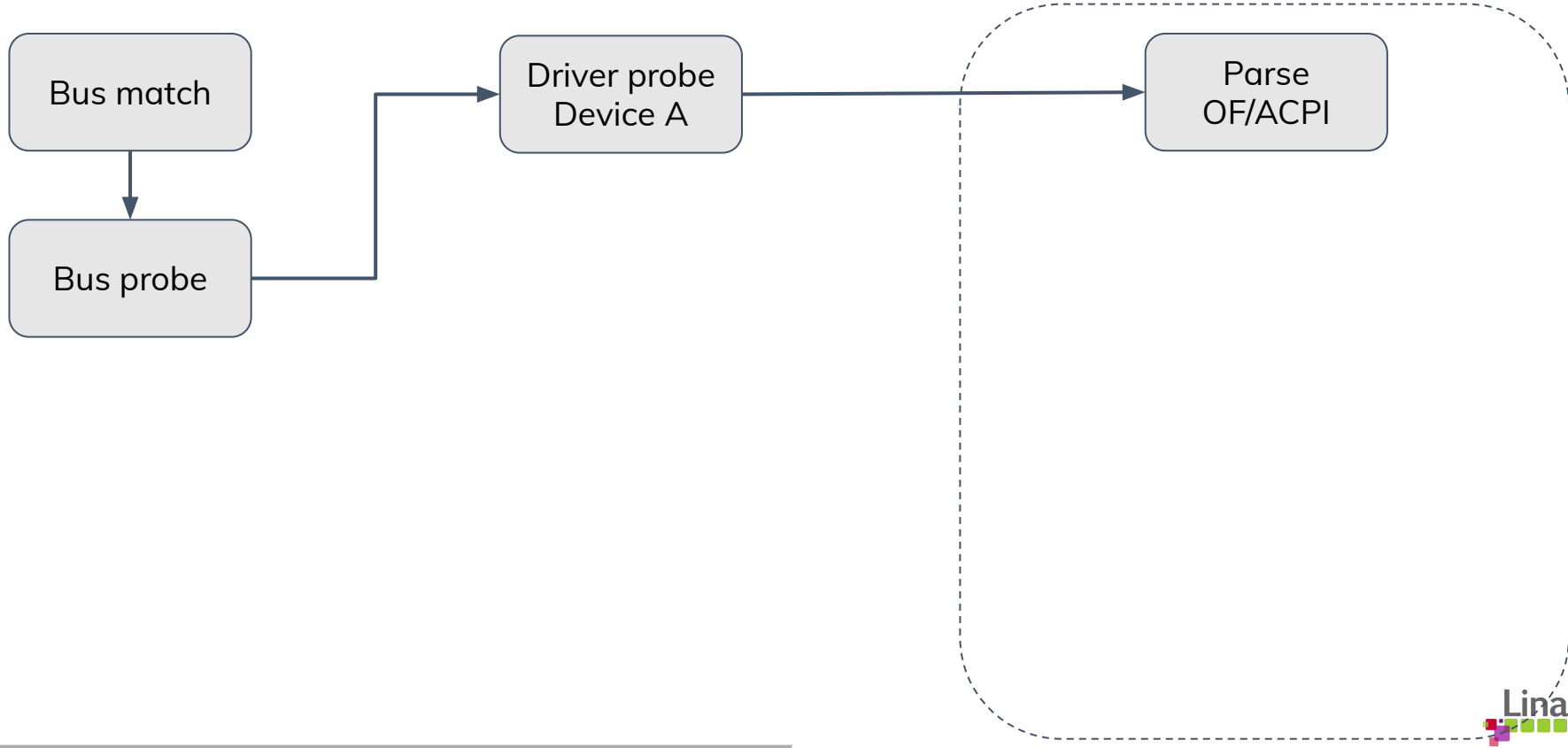

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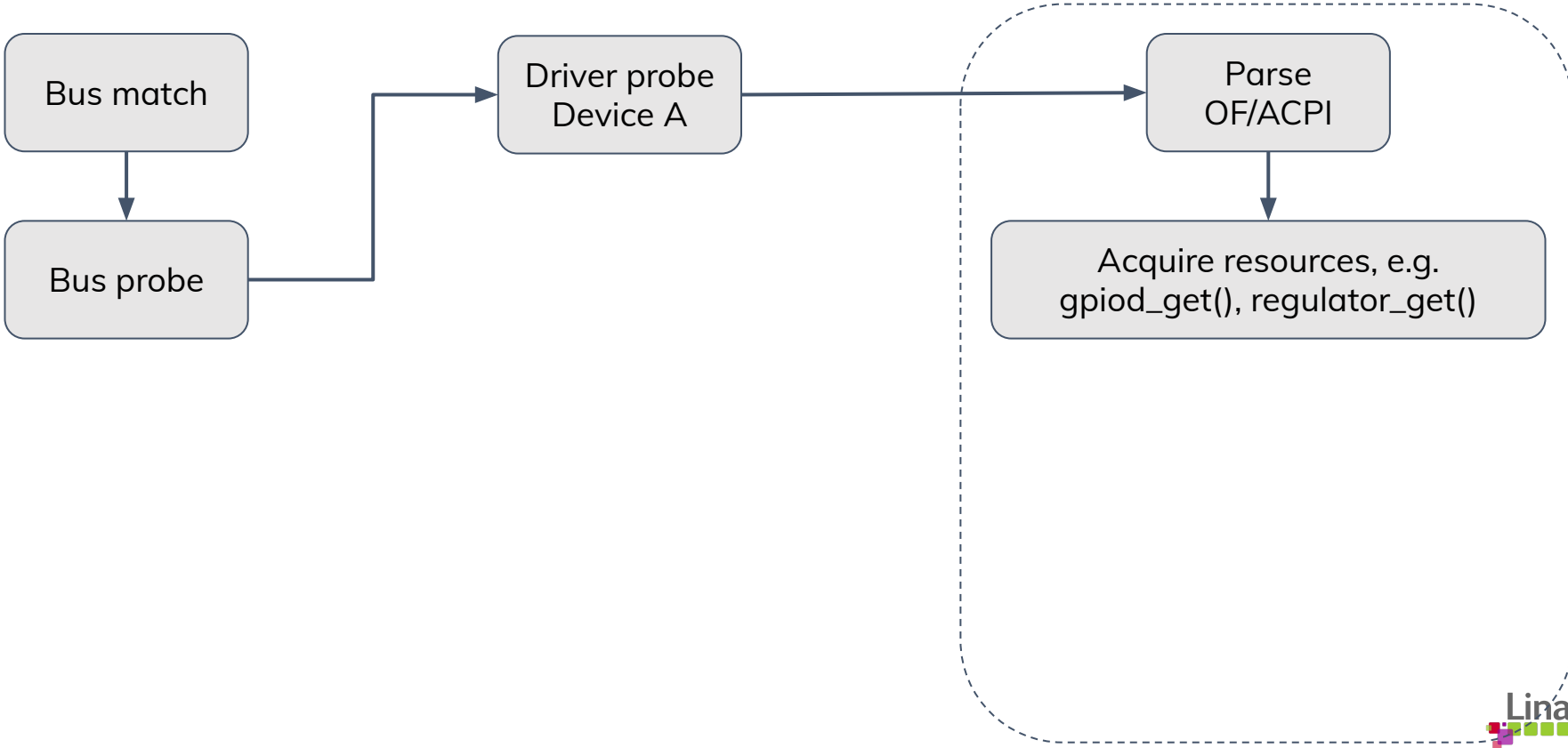

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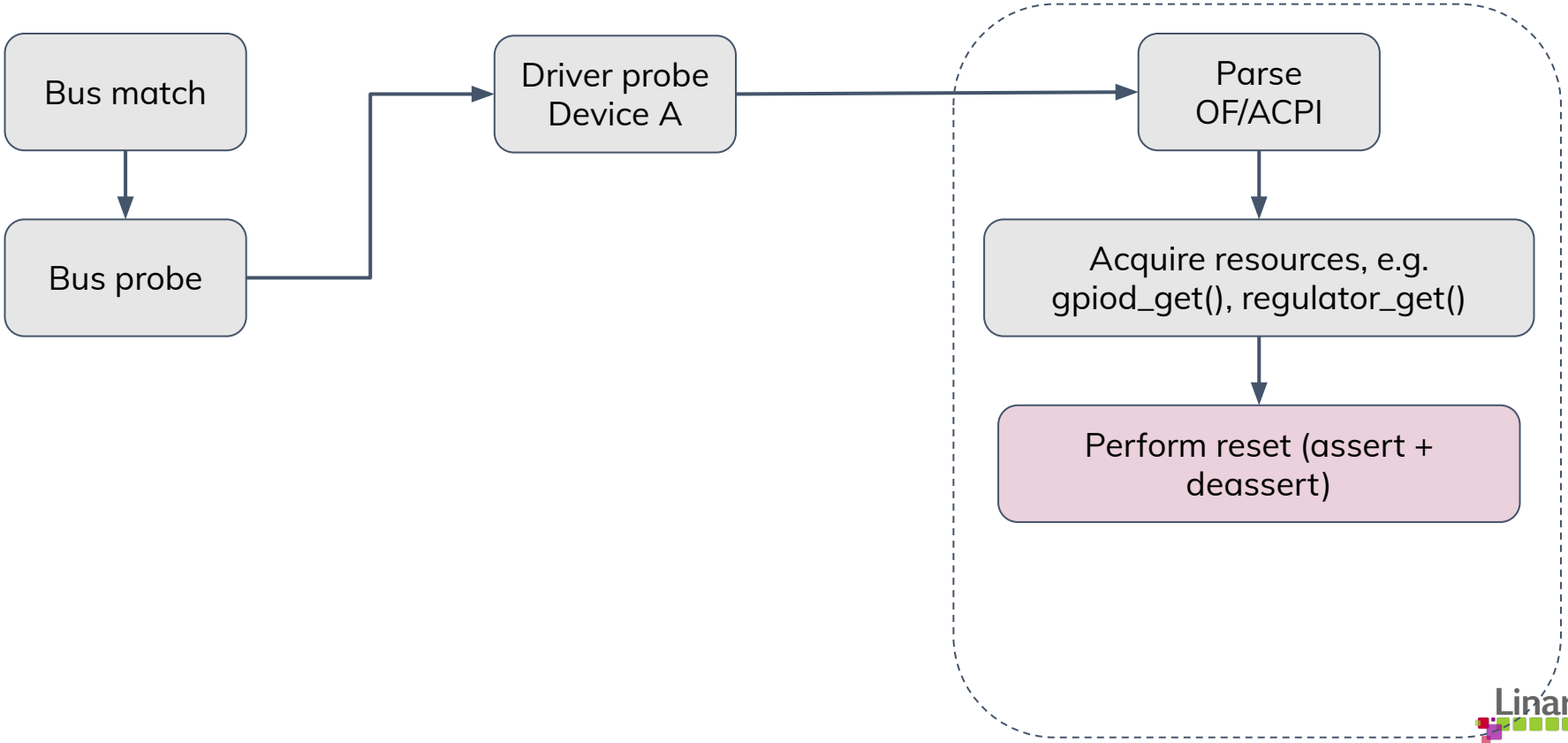

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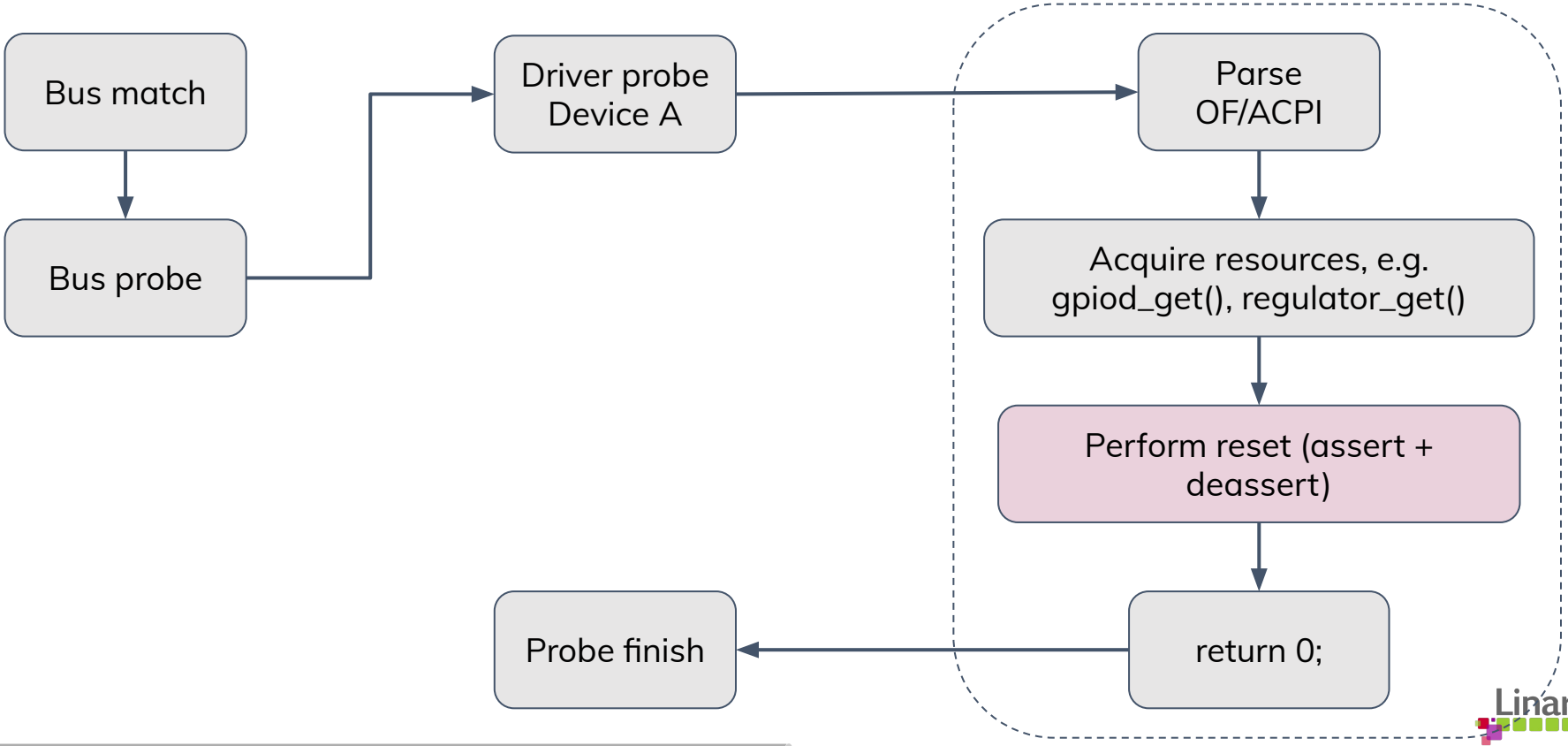

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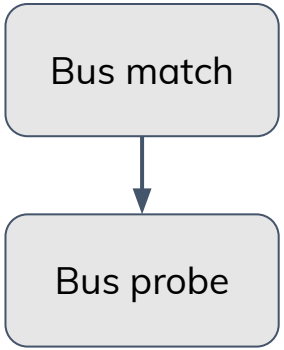

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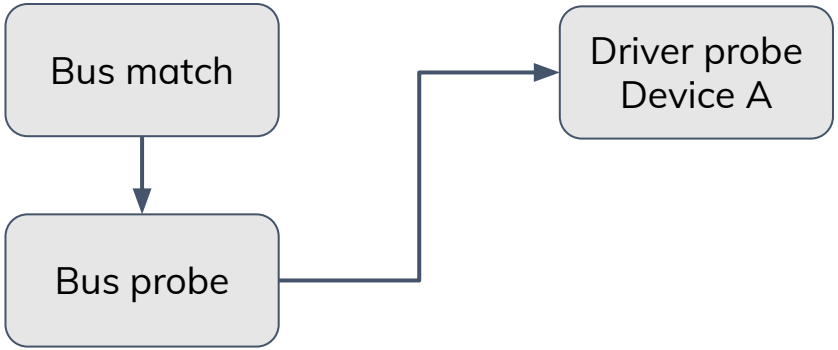

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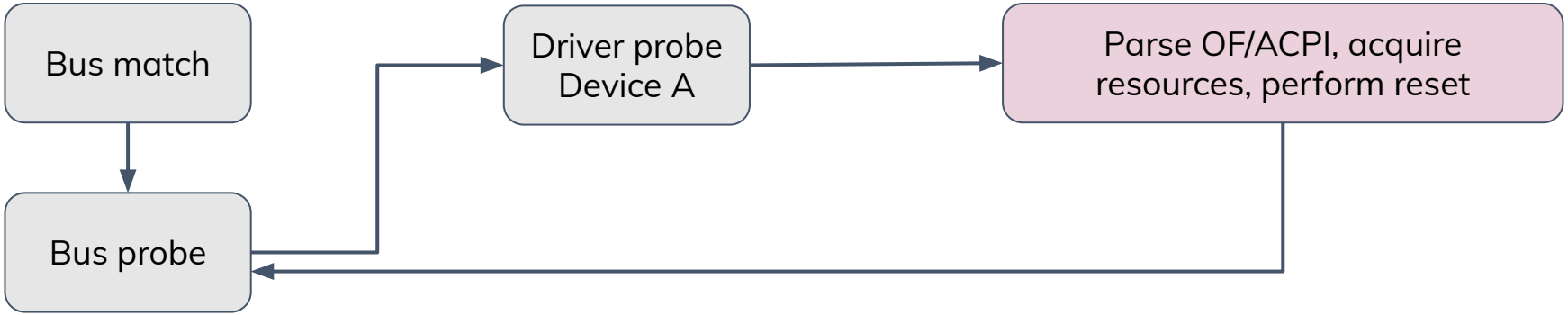

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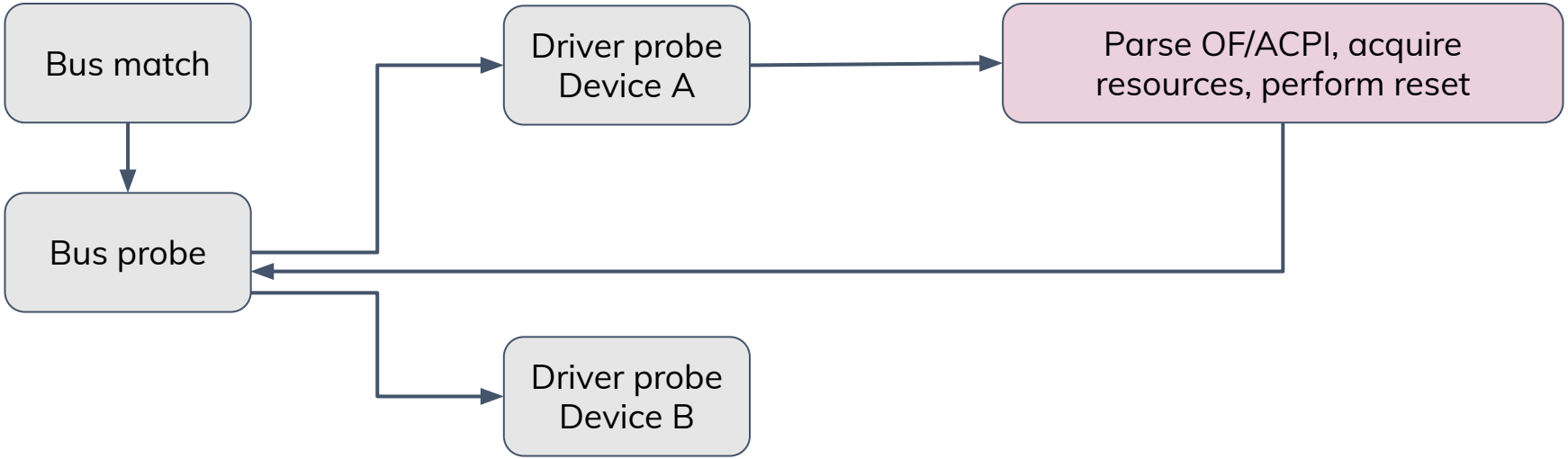

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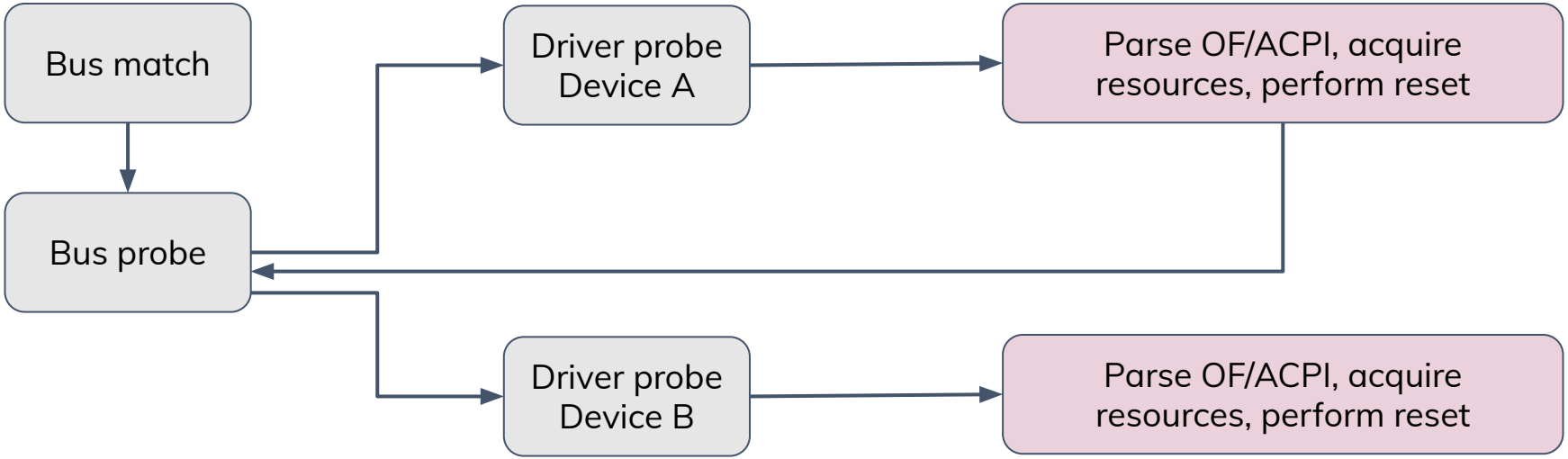

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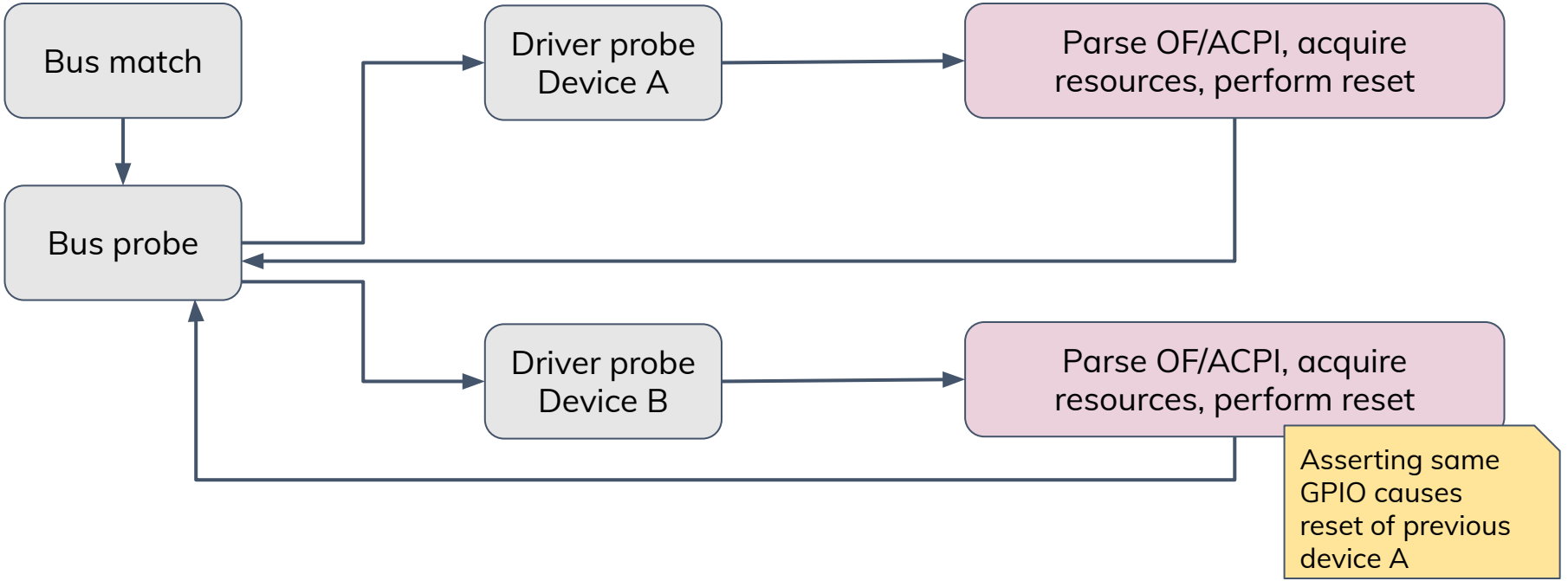

Solution: Non-exclusive GPIO

Requesting GPIO Twice

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- Two Linux devices requesting same GPIO:

```
wsa884x-codec sdw:1:0:0217:0204:00:0: error -EBUSY: Cannot get GPIO
```

- GPIO framework by default does not handle multiple users (drivers) of the same GPIO

Requesting GPIO Twice: Non-exclusive

- But there is a solution: GPIOD_FLAGS_BIT_NONEXCLUSIVE

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sd_n = devm_gpiod_get_optional(dev, "powerdown",
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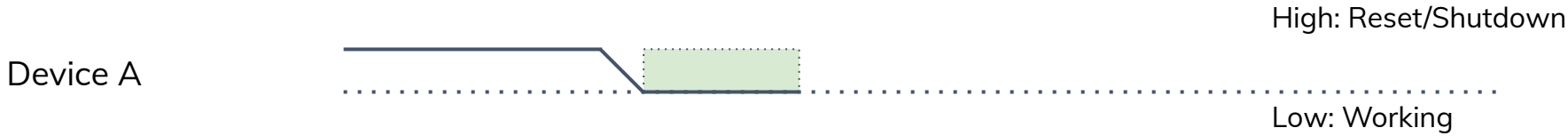
  - Does not solve problem of one device resetting the other one

Dumb Drivers with FLAGS_BIT_NONEXCLUSIVE



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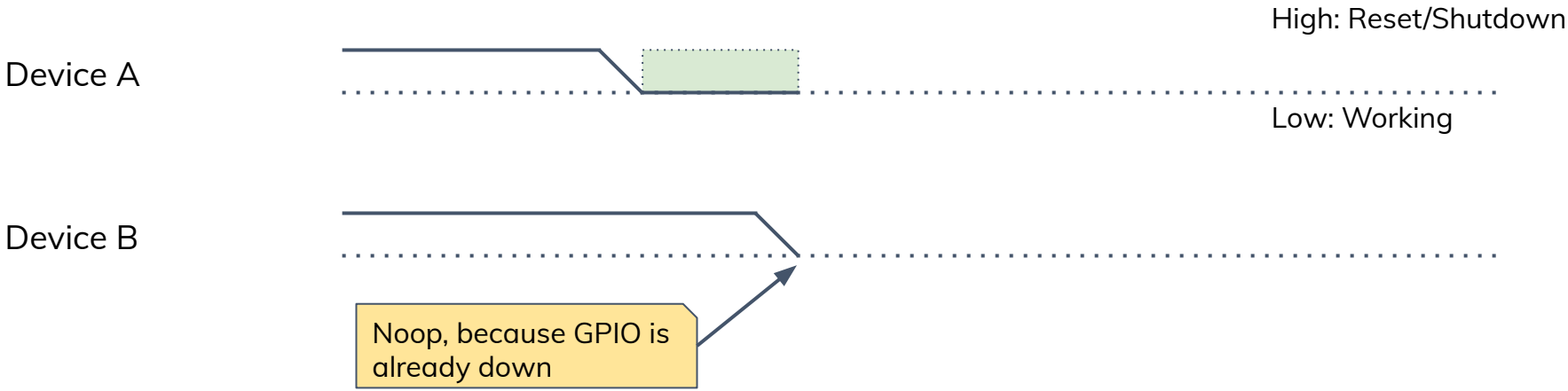

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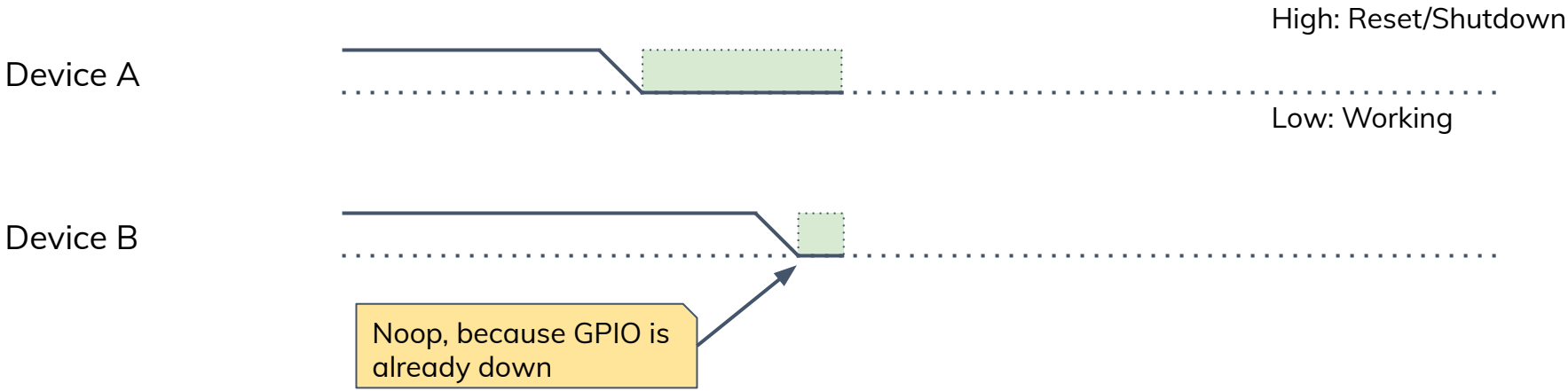

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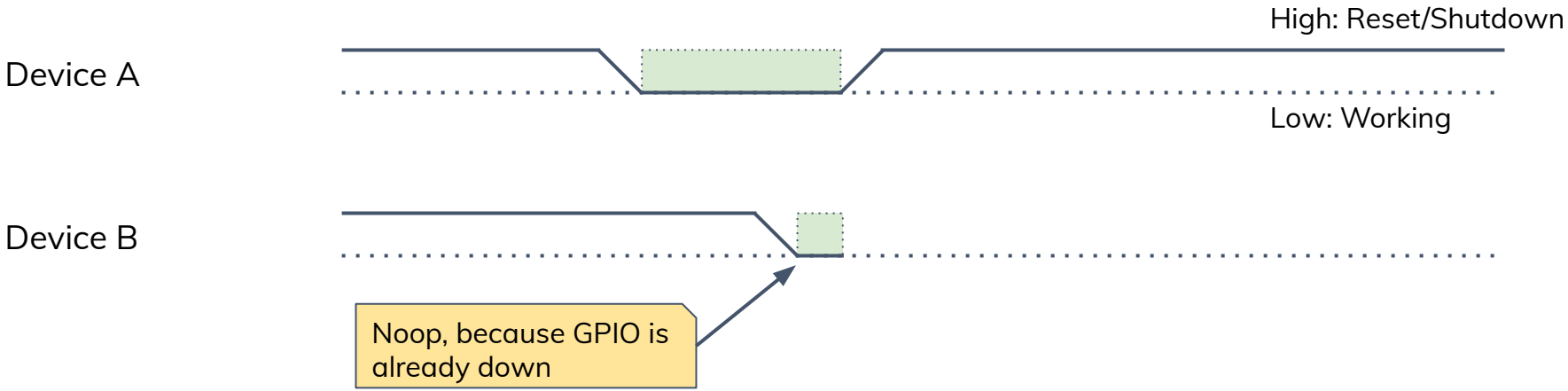

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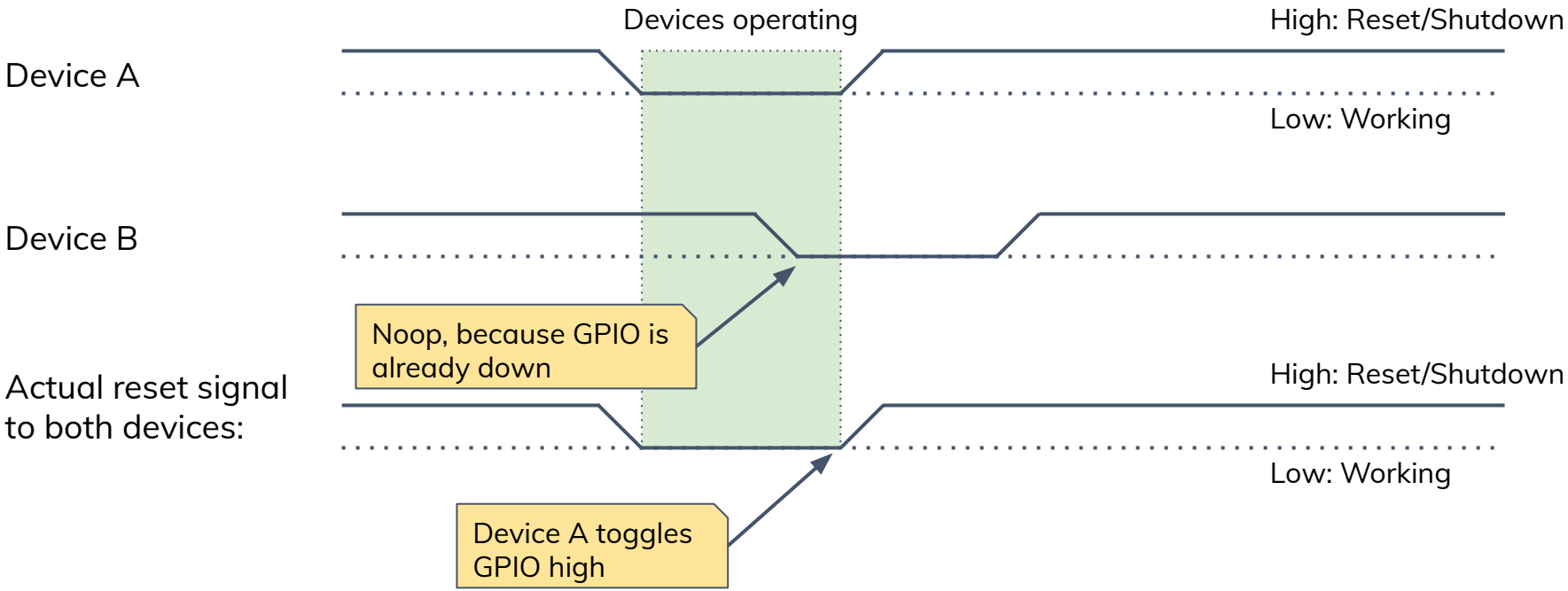

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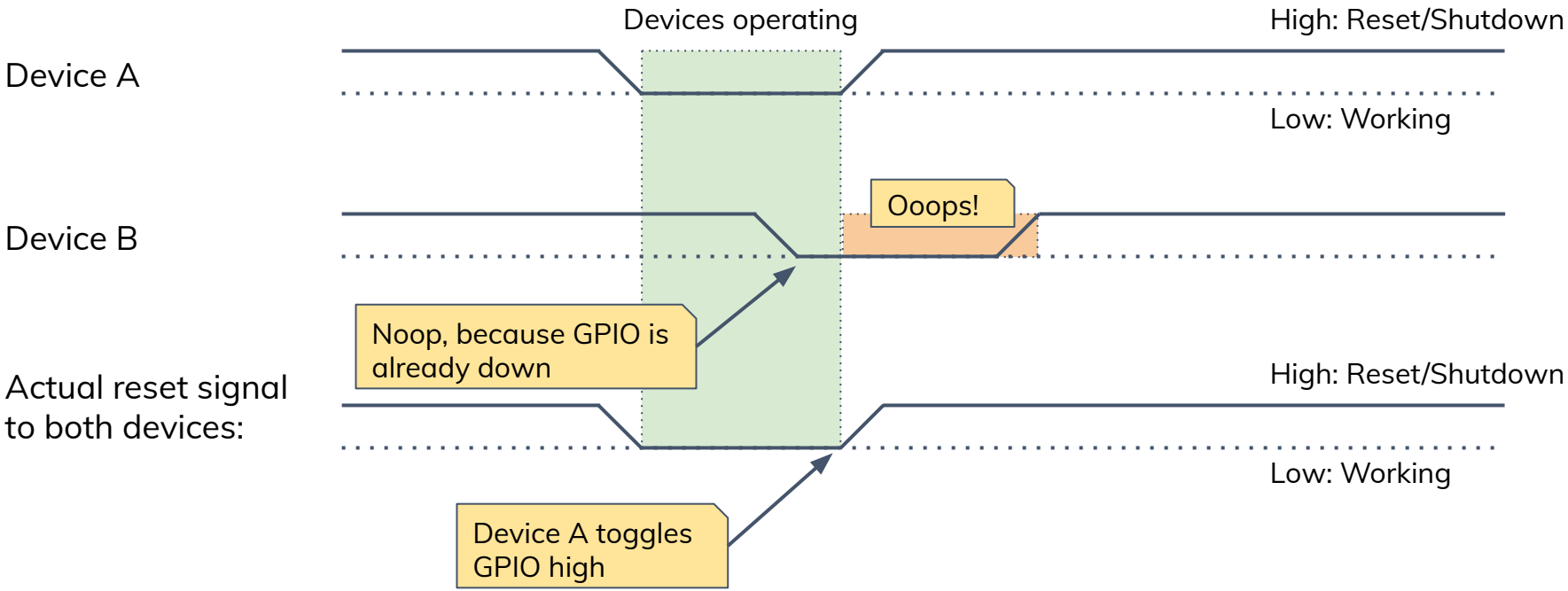

Linaro

Regulators

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- GPIOD_FLAGS_BIT_NONEXCLUSIVE was added for the Regulator framework
- Regulators handle it internally, thus using the same GPIO is perfectly fine

```
regulator-1 {
        compatible = "regulator-fixed";
        regulator-name = "foo-1";
        gpios = <&tlmm 17 GPIO_ACTIVE_HIGH>;
        enable-active-high;
};

regulator-2 {
        compatible = "regulator-fixed";
        regulator-name = "foo-2";
        gpios = <&tlmm 17 GPIO_ACTIVE_HIGH>;
        enable-active-high;
};
```

Solution: Reset Framework (v6.9-rc1)

Coming Soon with v6.9: reset-gpio

- Linux v6.9-rc1 merged my work of using Reset framework to handle reset GPIOs seamlessly
  - For shared reset/shutdown lines

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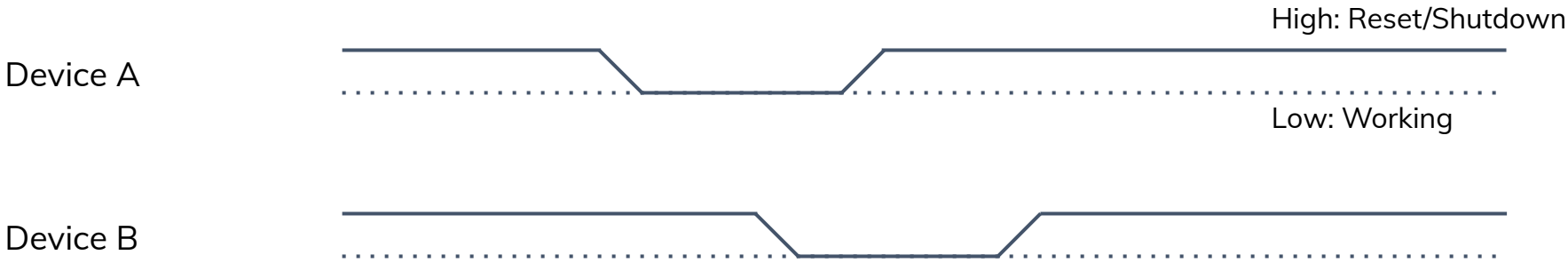
- Limited only to Devicetree platforms
- Examples:
  - drivers/i2c/muxes/i2c-mux-pca954x.c
  - sound/soc/codecs/wsa884x.c

Shared Reset or Shutdown Line, Active High



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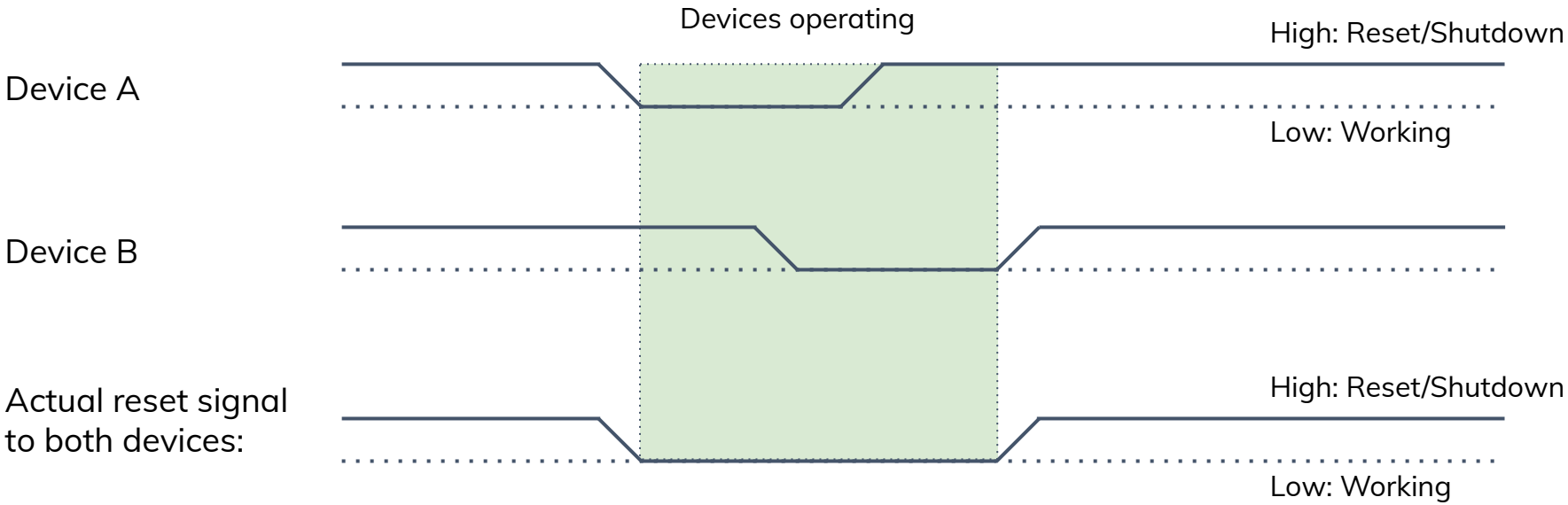

reset-gpio: Devicetree

```
      left_tweeter: speaker@0,1 {
            compatible = "sdw20217020400";
            reg = <0 1>;
-           powerdown-gpios = <&lpass_tlmm 12 GPIO_ACTIVE_LOW>;
+           reset-gpios = <&lpass_tlmm 12 GPIO_ACTIVE_LOW>;
            #sound-dai-cells = <0>;
            sound-name-prefix = "TwitterLeft";
            vdd-1p8-supply = <&vreg_l15b_1p8>;
```

No “resets” property

reset-gpio: Driver Probe

```
static int get_reset(struct device *dev, ...)
{
      ...









      sd_n = devm_gpiod_get_optional(dev, "powerdown", GPIOD_OUT_HIGH);
      if (IS_ERR(sd_n))
            return dev_err_probe(dev, PTR_ERR(sd_n),
                                 "Shutdown Control GPIO not found\n");

      return 0
};
```

reset-gpio: Driver Probe

```
 static int get_reset(struct device *dev, ...)
 {
       ...

+      sd_reset = devm_reset_control_get_optional_shared(dev, NULL);
+      if (IS_ERR(sd_reset))
+            return dev_err_probe(dev, PTR_ERR(sd_reset), "Failed to get reset\n");
+      else if (sd_reset)
+            return 0;
+      /*
+       * else: NULL, so use the backwards compatible way for powerdown-gpios,
+       * which does not handle sharing GPIO properly.
+       */
       sd_n = devm_gpiod_get_optional(dev, "powerdown", GPIOD_OUT_HIGH);
       if (IS_ERR(sd_n))
             return dev_err_probe(dev, PTR_ERR(sd_n),
                                  "Shutdown Control GPIO not found\n");

       return 0
 };
```

A shared reset!

reset-gpio: Driver Reset (De)Assert

```
 static void device_powerdown(void *data)
 {
       struct foo_priv *foo = data;

-      gpiod_direction_output(foo->sd_n, 1);
+      if (foo->sd_reset)
+            reset_control_assert(foo->sd_reset);
+      else
+            gpiod_direction_output(foo->sd_n, 1);
 }
```

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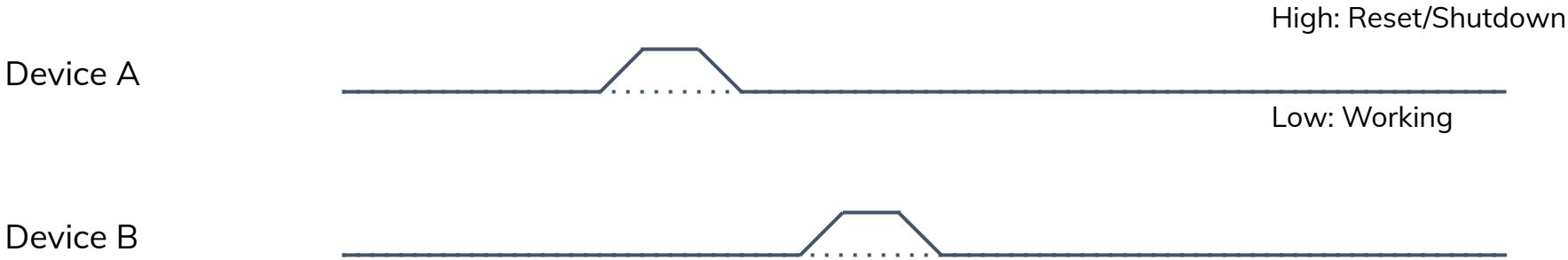

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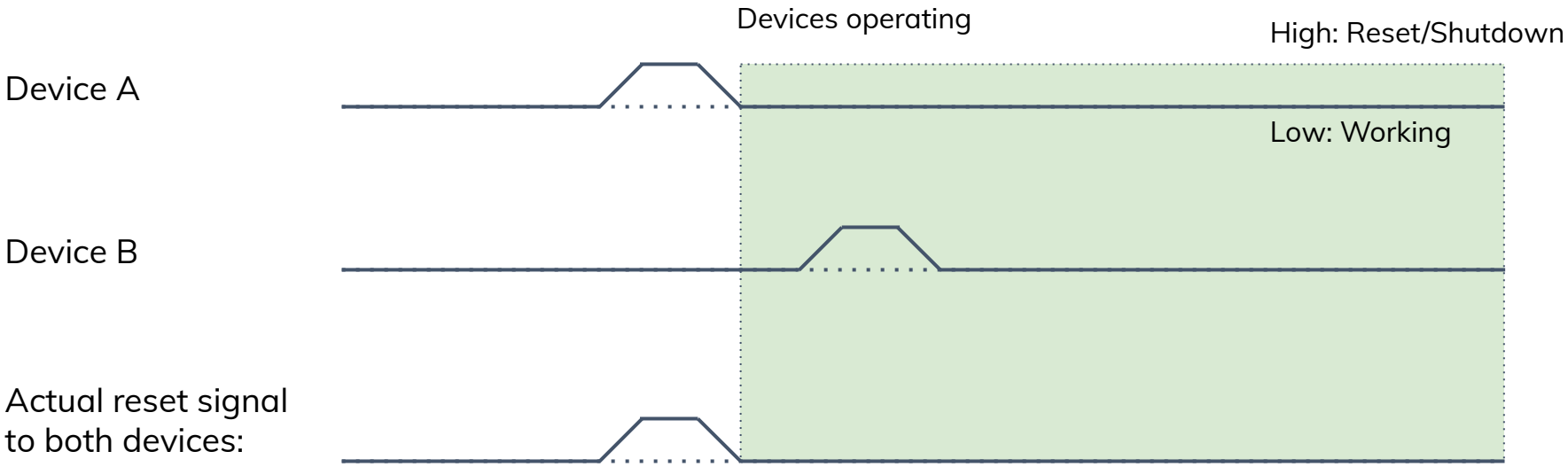

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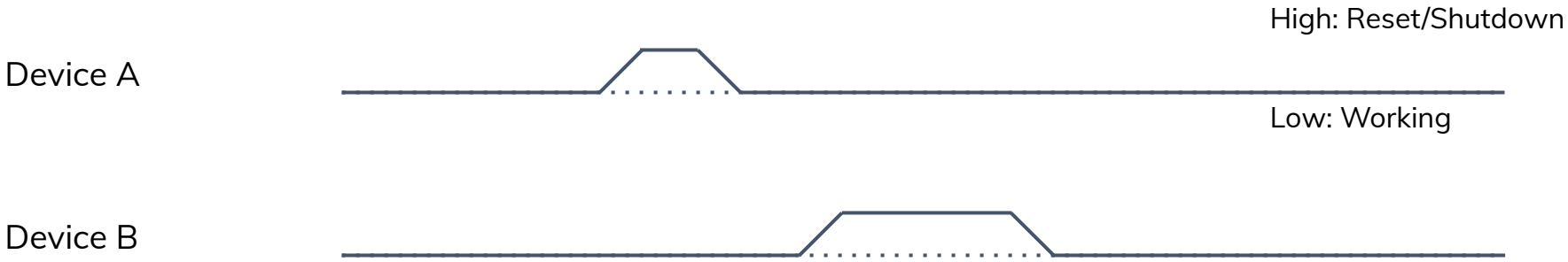
  - Different devices might have different requirement of pulse-length
  - Support for this could be added
    - See also previous work of Sean Anderson

Shared Reset Pulse with Different Pulse Length



Shared Reset Pulse with Different Pulse Length

High: Reset/Shutdown

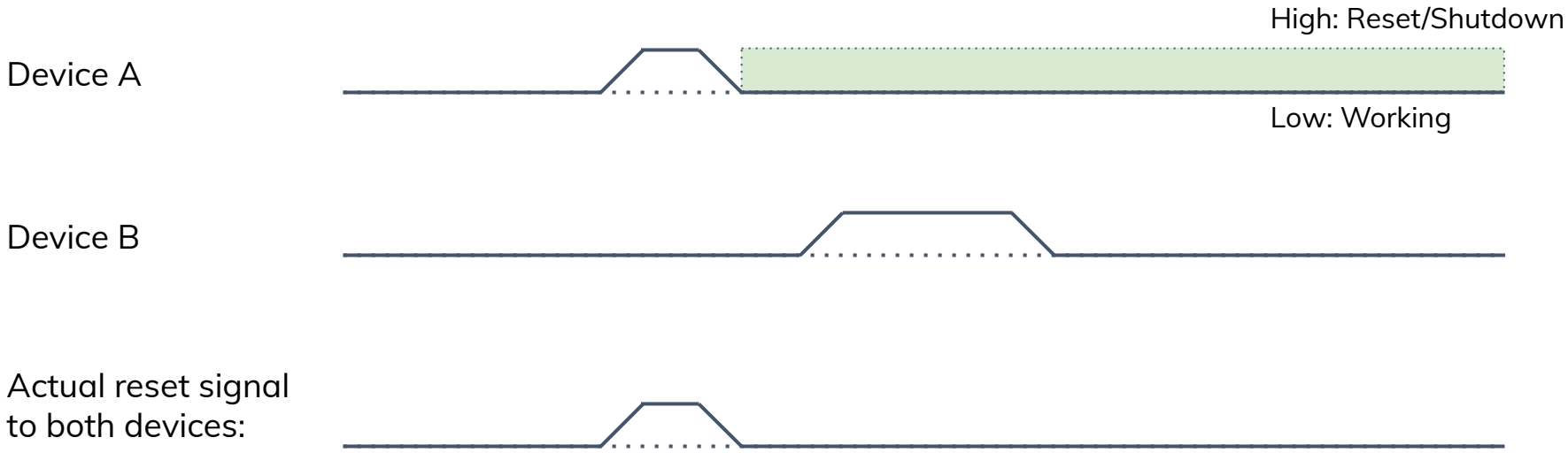
Device A

Low: Working

Device B

Actual reset signal
to both devices:

Shared Reset Pulse with Different Pulse Length



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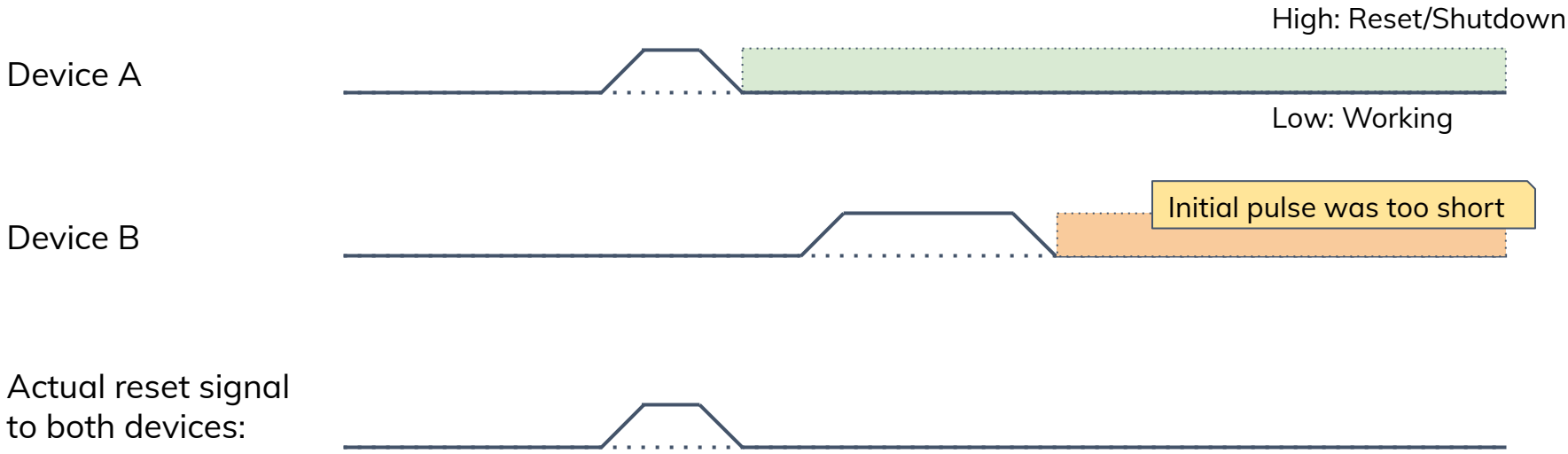

GPIO Aggregator and Delay

Linaro

GPIO Aggregator

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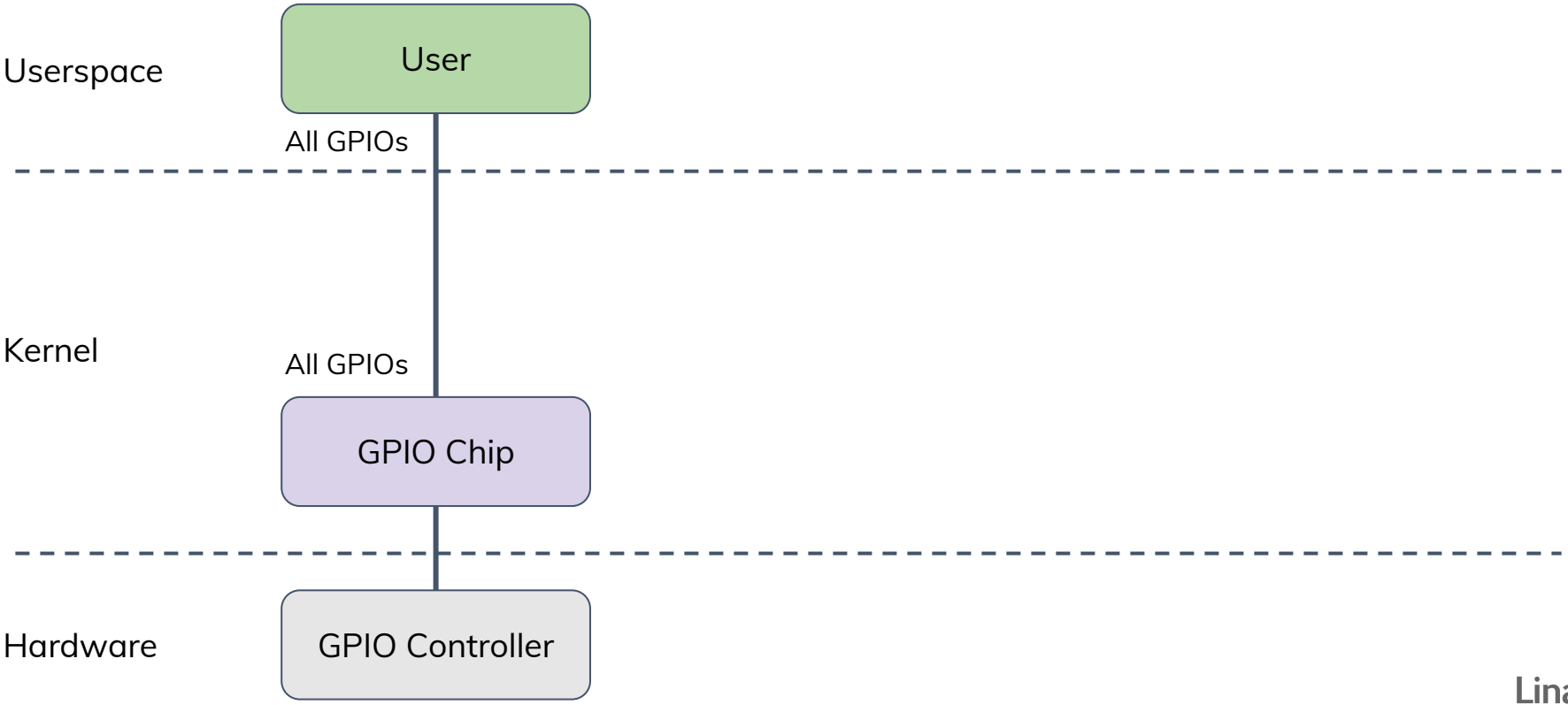
GPIO Aggregator: Access Control Example



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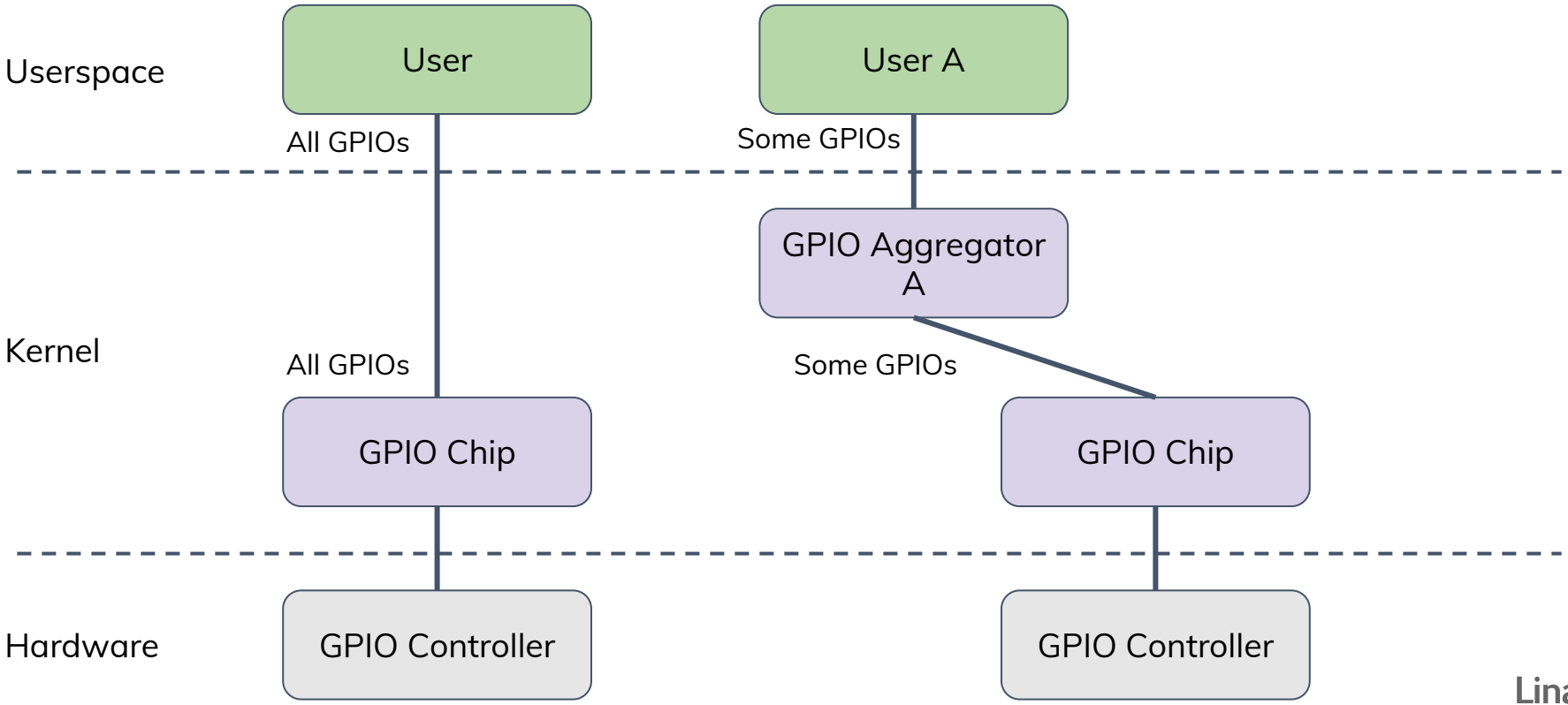

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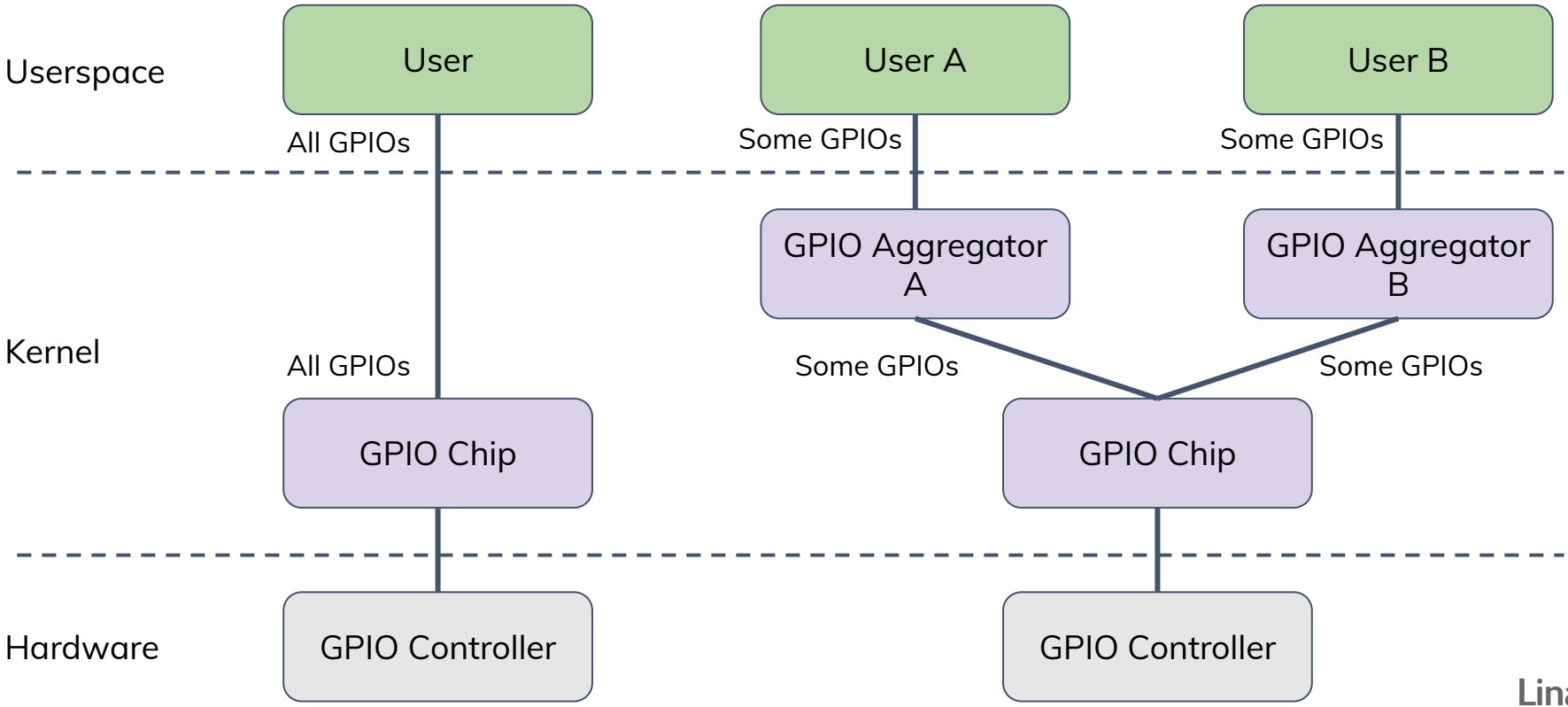

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    - GPIO aggregator allows to partition one GPIO controller into multiple controllers
    - Which in turn can be individually allowed per user or virtual machine
  - Generic GPIO Driver
    - Industrial control, where it can provide userspace access to a simple GPIO-operated device described in Devicetree (like spidev does for SPI-operated devices)

GPIO Aggregator and Delay

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  - Purpose: GPIO output delays introduced by RC circuits
- Currently both the driver and bindings support only 1-to-1 mapping
- With a some effort driver and bindings could support 1-to-many or many-to-1 mappings
- That’s just an idea…

Unresolved Problems

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  - Devicetree: Passing enable GPIOs as “reset-gpios” property, to use new reset-gpio Linux driver, would not be accepted by Devicetree maintainer

Other Unresolved Problems

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  - Reset pulses with reset-gpio driver

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  - Different delays for reset pulses
- Devices needing to do something immediately after the reset
  - Not even sure how such driver could work, considering modules and deferred probe
  - Blame hardware engineers...

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